

Measuring Individual Video QoE: A Survey, and Proposal for Future Directions Using Social Media

YI ZHU, Delft University of Technology

SHARATH CHANDRA GUNTUKU, University of Pennsylvania

WEISI LIN, Nanyang Technological University

GHEORGHITA GHINEA, Brunel University

JUDITH A. REDI, Delft University of Technology

The next generation of multimedia services have to be optimized in a personalized way, taking user factors into account for the evaluation of individual experience. Previous works have investigated the influence of user factors mostly in a controlled laboratory environment which often includes a limited number of users and fails to reflect real-life environment. Social media, especially Facebook, provide an interesting alternative for Internet-based subjective evaluation. In this article, we develop (and open-source) a Facebook application, named YouQ¹, as an experimental platform for studying individual experience for videos. Our results show that subjective experiments based on YouQ can produce reliable results as compared to a controlled laboratory experiment. Additionally, YouQ has the ability to collect user information automatically from Facebook, which can be used for modeling individual experience.

CCS Concepts: • **Information systems** → **Social networks**; • **Social and professional topics** → **User characteristics**; • **Human-centered computing** → *Social networks*;

Additional Key Words and Phrases: Individual Quality of Experience, user factors, Facebook

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1 INTRODUCTION

Internet-based multimedia fruition has exploded in the recent years. According to Cisco's forecasts, video delivery will account for 80% of the overall Internet consumer traffic by 2019 [1]. Newer and more immersive formats are becoming more common; high-frame-rate, panoramic, HDR video is on the verge of becoming regular (live) streamed content. Content delivery systems now face the

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Authors' addresses: Y. Zhu and J. A. Redi, Delft University of Technology, PO Box 5031, 2600 GA Delft, The Netherlands; emails: {Y.Zhu-1, J.A.Redi}@tudelft.nl; S. C. Guntuku, University of Pennsylvania, Philadelphia, PA, 19104, USA; email: sharathg@sas.upenn.edu; W. Lin, Nanyang Technological University, 50 Nanyang Avenue, 639798, Singapore; email: wslin@ntu.edu.sg; G. Ghinea, Brunel University, Uxbridge, Middlesex, UB8 3PH, United Kingdom; email: george.ghinea@brunel.ac.uk.

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challenge of meeting users' expectations in terms of quality of multimedia fruition while facing the current infrastructural limitations. User tolerance for impairments and buffering events in video playback is decreasing exponentially, and influences users' choice in adopting one service or content provider over the other [2]. On the other hand, resource scarcity (e.g., bandwidth) makes flawless delivery more challenging, making it necessary for a new generation of smart resource optimizing and delivery mechanisms. To do that, it is essential to devise accurate tools to measure the extent to which the user deems the multimedia experience to be of a high quality.

The concept of Quality of Multimedia Experience (QoE) has been shaped to address this necessity. According to the Qualinet White Paper [3], "Quality of Experience is the degree of delight or annoyance of the user of an application or service. It results from the fulfilment of his or her expectations with respect to the utility and/or enjoyment of the application or service in the light of the user's personality and current state". As such, QoE is determined by the characteristics of the multimedia system that delivers the experience, as well as the context in which the experience is delivered. However, multimedia experiences are also, and more importantly, individual and their appreciation and durability differs from user to user [4, 5].

Traditionally, multimedia delivery has been optimized for the 'average user', and QoE measured through so-called Mean Opinion Scores [6]. Multimedia quality metrics, aimed at measuring QoE based on the analysis of either the encoded or decoded media stream [7], have been designed to predict the media quality as perceived by such an average user. Individual preferences have mostly been neglected, due to the inherent difficulty in dealing with quantifying and measuring individual characteristics.

However, in reality, there is no such thing as an 'average user'. We see this in (computational) advertising, for example: the same product is advertised differently on different markets according to their demographics; products are branded differently to take into account local sensibilities or preferences. In the domain of (multimedia) recommendation [8, 81], individual preferences and diversity [9] are taken into account. Personalized solutions for education, conservation, and (e-) health, UX, to name but a few domains, are becoming the rule, rather than the exception [10, 82].

Similarly, perceived QoE varies from user to user. Recently, several studies have examined the influence of user factors on individual QoE: users with different age, gender, personal interests, and personality experience the same multimedia differently [11, 12, 74, 75, 76, 77]. Research has also shown that by incorporating user information into quality metrics, reliable estimations of individual QoE can be obtained [4], and user-based (over and above content-based) delivery optimization is within reach.

QoE, by definition, is supposed to be subjective and individual. However, we use the term "individual QoE" since the majority of the literature on QoE has not treated it as such. Whilst the literature conveniently averages QoE (just like with the average family having 2.4 children), in reality averaging ignores the fact that QoE is unique to each individual and to each individual's experience and that it could depend (or not, as the context may be) on an individual's cognitive style, personality, learning style, affective state, psychophysical acuity, as well as on any of a number environmental factors such as lighting, ambient noise, access device, not to mention task being undertaken. The challenge is that the set of individual factors upon which an individual's QoE depends is not fixed; rather this (sub)set varies from one context to another, and it is this what justifies even more emphatically the individuality and uniqueness of a user's experience – hence the term "individual QoE."

Despite encouraging initial results, research on individual QoE is still in its infancy. The number of and the extent to which user factors may influence QoE is enormous, and a collaborative, community-wide effort is needed to test several user factors systematically. In fact, in order to get reliable results, researchers usually focus on one (or few) specific user factors (e.g., interest [11],

personality [11, 12, 78]), following standard protocols and methods (i.e., ITU-R BT500 [6]) in a controlled laboratory environment. However, there are several other factors beyond the laboratory environment that may potentially influence multimedia experiences. Additionally, when it comes to individual QoE evaluation, the studies in the laboratory often include limited number of users and fail to reflect real-life consumption scenarios [13].

Lately the scientific community has devoted increasing attention to the usage of web and Internet platforms to perform user testing [14–17]. Internet-based subjective experimentation refers to all types of experiments conducted through the Internet (e.g., e-mail, websites, crowdsourcing platforms, or social media) [15]. It allows reaching users all over the world (with consequent diversity in demographics and cultural backgrounds) and in ecologically valid consumption conditions. For example, millions of users answered a personality questionnaire, which was traditionally used in the laboratory [15].

The QoE community has followed this trend, focusing especially on crowdsourcing [18] as a method to collect subjective QoE evaluations in a fast and cost-effective way. By posting short QoE evaluation tasks on crowdsourcing platforms (e.g., Mechanical Turk, Microworkers), researchers can reach out to a vast number of users across the world, who complete evaluations in a resource-effective manner (both in terms of time and money). A number of studies have shown that crowdsourced subjective testing can produce QoE assessments that are as reliable as those that would be obtained in a controlled laboratory setting [13, 19, 20]. However, they need to be short to let the user stay focused [14]. As a result, there is a limit to the number of factors which can be explored in each experiment.

Social media, especially Facebook, could serve as an interesting platform for Internet-based subjective experimentation. With over 1.65 billion monthly active users [17], Facebook has become a major platform for people to share their experiences and preferences (e.g., by expressing likes/dislikes or providing status updates) [21]. Facebook can be seen as a huge real-life, continuously growing database of human behavior. Evidence has shown that data collected from Facebook (e.g., status updates) reflects the user's actual ideas and feelings outside the social network [22, 23], and individual traits such as, e.g., personality and health, can be inferred through automatic analysis of the text of comments and posts [24, 76, 79, 80]. Different from other social media platforms, Facebook allows researchers to access user information by developing a Facebook application [17]. Through careful design, by embedding QoE subjective tests into Facebook applications, it is possible to reach potentially millions of users and collect a substantial dataset for research [17], where personal preferences and traits can also be taken into account and studied in relation to individual QoE appreciation. However, questions arise again with respect to the reliability of the data collected through such an application, since users of the application are often from an uncontrolled environment.

In this article, we also investigate the potential for Facebook as a reliable platform for studying individual QoE evaluations. Specifically, we are interested in answering the following question: *Can Internet-based subjective experimentation based on Facebook application produce reliable results for individual Quality of Experience evaluation for videos, as compared to a controlled laboratory experiment?*

To answer this question, we developed YouQ¹, an open-sourced Facebook app designed to perform subjective experiments aimed at measuring individual QoE while watching videos. To test its effectiveness, we performed an experiment investigating the impact of personality and demographics on the perceived QoE of videos with different genres and arousal levels. We performed this experiment both in the wild (i.e., openly on Facebook), and in a controlled lab setting to

¹YouQ: ii.tudelft.nl/qoe/FbApp/.

measure the difference that looser control and higher ecological validity bring to controlled QoE evaluations. In addition, data from our online experiment is further compared with two previous independent studies [11, 12], to verify whether results from our Facebook-based experiment are in line with the literature, and whether it can provide additional insights to the growing field of research addressing user factors in QoE. Further, we also provide a meta-analysis of the state-of-the-art literature studying the influence of user factors on multimedia experience.

2 RELATED WORK

2.1 Factors Influencing Multimedia Quality of Experience

In the 1990s, user satisfaction with multimedia systems and services was mainly linked to the concept of Quality of Service (QoS), defined as the “totality of characteristics of a telecommunication service that bears on its ability to satisfy stated and implied needs of the user of the service” [25]. In fact, QoS focuses mainly on quantifying system and network performance, based on metrics such as frame-rate, resolution, packet loss ratio, lag, and the like. These metrics, which purely describe the system functioning, were soon shown to correlate poorly with actual user satisfaction [26]. In view of this discrepancy, a more user-centric concept of Quality of Experience (QoE) started to emerge and moved the focus towards quantifying the quality of media as perceived by the user. Research in perceptual quality has seen a substantial amount of work in the last two decades [7], incorporating the influence of core user factors such as perception and attention in QoE optimization. On the other hand, most of these efforts were still devoted to maximize QoE for an “average user”, developing a single solution for any user adopting the system or service, irrespective of the users’ background.

Lately, the view of what QoE is and how it should be measured has evolved dramatically. The Qualinet White Paper on definitions of Quality of Experience has painted a new picture with respect to what influences QoE, and as such should be taken into account in multimedia delivery optimization [3]. In this new view, the QoE delivered by a system is not only influenced by the properties of the system itself (system factors) but also by the context in which the experience is consumed (context factors), and by the current state of the user, i.e., by the individual characteristics of the user.

2.1.1 System Factors. System factors determine “the technically produced quality of an application or service” [3]. They can influence QoE in video delivery services by altering the perceptual quality of the content. For example, visible artifacts such as blockiness, blur, and ringing might be generated by a certain type of compression (e.g., H.264/AVC). These can result in user dissatisfaction. Similarly, QoE can be impacted by network QoS parameters [27], and the media configuration [28]. For example, buffer ratio, i.e., the fraction of time spent in buffering over the total session time, including playing plus buffering, is inversely related to QoE [27]. Similar observations were made for other QoS parameters such as buffering events rate, buffering duration and average bitrate [29]. Besides these, the nature of video content itself can influence users’ QoE [30]. For example, different genres (e.g., comedy, action, etc.) have different viewing patterns, resulting in different perceptual quality. For instance, for a given bitrate, genres which contain little movement, e.g., comedy sequences, usually have higher perceptual quality than genres which contain high-pace movement, e.g., action sequences [31].

2.1.2 Context Factors. Context factors describe the situation in which the media is consumed by the user. They entail characteristics of the physical environment, the presence (*in situ* or remote) of other users experiencing the same media, or economic conditions regulating for the service fruition. Physical surroundings that may influence users’ experience are the seating position – viewing distance and height, lighting conditions [32] and disturbances which might occur such as

incoming phone calls or SMS message alerts [33]. Co-viewing videos with friends may make the entire experience more enjoyable, and therefore of higher quality [11]. Finally, depending on the amount of money paid for accessing the content, users may have different expectations in terms of quality, with lower prices corresponding to higher tolerance to impairments [34].

2.2 Preliminary Studies on User Factors and Individual Differences in QoE

Differential Psychology has been studying individual differences in user factors and their impact on human behavior and performance for over a century. Despite this large body of knowledge, individual differences and user factors have mostly been neglected in multimedia delivery optimization, with the exception of a few studies. Their origin is scattered, coming from the fields of Media Psychology and Human-Computer Interaction, and only lately from the Multimedia community. Although the literature in this area is far too thin to build a complete taxonomy of the effect of individual differences and user factors on Quality of Experience, we can identify four macro-categories within which they can be clustered: demographic, physiological, psychological, and socio-cultural factors.

2.2.1 Physiological Factors. Because of the configuration of the human sensory system, physiological characteristics of the individual users play a crucial role in QoE. As such, they have been thoroughly investigated, and models of visual perception and attention have been incorporated in objective video quality assessment metrics. Nevertheless, these factors have been mostly studied and modeled referring, once again, to the “average user”. Fewer studies have looked into how individual physiological characteristics influence QoE. Elements such as visual acuity and losses in contrast sensitivity due to aging have been shown to influence visibility (and annoyance) of visual impairments; yet, they have hardly been included in QoE models [35]. Color blindness also alters users’ perception and, when it comes to immersive, 3D visual representations, stereo-blindness can be a core factor in determining the appreciation of new imaging technologies [36]. Similarly, human hearing characteristics can influence QoE with auditory (or audiovisual) media; in fact, international recommendations for subjective testing conditions provide a set of indications to minimize individual differences in QoE ratings due to physiological diversity, rather than acknowledging them [37].

2.2.2 Socio-Cultural Factors. Both the educational and the socio-cultural background of the user play important roles in experiencing media. Cultural background has been shown to influence visual perception and attention, for example, between Westerners and Asians in [38]: Americans were reported to have more analytical visual perception (inclined to pay attention to details), while Asians were seen to have a more holistic visual perception (likely to be more sensitive to context). The influence of culture on optical illusion, color perception, visual attention, and brain functioning was documented in [39]. The correlation between culture and cognition was also studied in [40], by analyzing the variation of word associations given by Japanese and American participants. Previous experience with multimedia technology and devices is also critical in determining one’s QoE. Users with expertise in photography were shown to be more consistent than naïve users in judging the aesthetic appeal of images; users less acquainted with mobile media consumption were found to be more critical and to have different expectations, in terms of QoE, with respect to other users who were more technologically savvy [5, 13].

2.2.3 Demographics. Demography (e.g., age or biological sex, nationality) may influence QoE. For instance, older adults are found to be more critical than younger users suggesting that elderly people usually have higher requirements for QoE [41]. On the contrary, another study showed

that younger users tend to rate video quality lower than older users, although the performance is better [42]. Similar scattered results exist for biological sex [5, 13], and no uniform conclusion has been carried out, to the best of our knowledge, to clarify the role of demographic characteristics in QoE appreciation.

2.2.4 Psychological Factors. The affective state of the user plays most likely the biggest role in determining one's satisfaction with the delivered experience. The user's mood has been shown to influence quality preferences, and multimedia experiences have been shown to influence mood in turn [13]. Among emotions, the most impactful one for QoE has been shown to be interest [43]. A number of studies indicate that QoE is triggered when something resonates with a user's interest [44] and that personal interest in video content significantly influences user's QoE judgment [45]. The more a user is interested in the video content, the higher QoE she/he got with the same bitrate [46]: football fans have been shown to be able to tolerate incredibly low frame-rates, when watching a match they like. Finally, user performance is proved to be influenced by personality. For example, enthusiastic people more likely to switch the TV channel or change the volume of the TV on their first attempt compared neurotic people [41]. Researchers found that including user expectations, users' monetary budgets, and quality pricing in modeling perceptual quality evaluation leads to increased accuracy [47–49].

Further, there have been other works which studied the variability of QoE based on other factors such as the device used for streaming the content [50]. While we tried to present most of the factors in this section, this is not an exhaustive list.

2.3 Measuring QoE: From the Laboratory to the Real-world

Subjective QoE evaluation is based on self-report ratings of one's (perceptual) satisfaction with respect to multimedia contents. Subjective ratings are often collected via psychometric experiments in the lab [51] and aim at measuring the satisfaction of an average user regarding a given video. To ensure reliability and repeatability of the experimental results, the community has developed standardized methodologies and recommendations on experimental settings for subjective testing [6]. These recommendations often include indications on the setup of the physical surroundings, assuming experiments are performed in a controlled laboratory environment for increased control over the execution of the experimental task.

Although ensuring high reliability and repeatability, lab-based experimentation also presents several inconveniences: (1) being highly time-consuming, the number of conditions (in our case, combinations of user and system factors) is necessarily limited, which typically leads to investigating at most one or two user factors at a time; (2) the demography of the users is often not representative of the general population, as most laboratories are embedded in technical faculties of universities, and most participants are students; and (3) the lab environment often fails to simulate the real-life environment where multimedia is consumed [13], thereby hampering the ecological validity of the results.

A growing number of researchers are now attempting to overcome the above inconveniences by conducting Internet-based subjective experiments (e.g., via websites, social media, and/or crowdsourcing platforms) [20]. Internet-based experimentations enable new possibilities for individual QoE evaluation, by allowing researchers to recruit users from a larger, more diverse group and users to perform the experiment in their real-life conditions, leading to more representative results [13]. In fact, psychologists have used the Internet to study human behavior for a decade [15]. The most straightforward application of online experiment is implementing traditional lab experiment online [16], i.e., using Internet technology (e.g., E-mail, website) to deliver surveys or questionnaires [15]. Very successful examples are the outofservice.com website and the Project Implicit

website², which both host standard personality questionnaires [52], and have reached millions of users [15].

Within QoE evaluation research, it is not straightforward to move from laboratory experiment to online experiment by simply generating a web version of the existing lab test [14], because of conceptual, technical, and motivational differences [53]. Conceptual differences are due to the fact that online experiments usually should be much shorter than a laboratory experiment, lasting at most 10 minutes to prevent the participant from losing focus and interest (where lab-based experiments can last more than 1 hour) [13]. The online experiment usually needs to be split into multiple smaller tasks, orchestrated to cover the entire condition set and avoid context effects [12]. Moreover, the online experiment is less supervised as compared to the lab experiment because it is not possible to give direct and instant feedback between users and supervisors during the experiment [13]. This implies that Internet-based experiments need to be carefully designed in order to ensure that each user understands the task without any difficulties (by providing a proper training session) [13]. The technical differences related to the web-based nature of the online experiment is that it is performed in a more real-life environment using personal device in contrast to the standardized environment and device in the lab-based evaluation [12]. This implies that advanced experimental devices (e.g., eye tracking) are not available in the online environment. In addition, the fact that the stimuli (e.g., test videos) has to be delivered over possibly slow connections to the user has to be considered. Finally, motivational differences have to be taken into account. Whereas in the lab experiment participants are motivated by genuine interest in the research and/or by social pressure, in Internet-based experiments extrinsic motivation is to be generated via either financial incentives (i.e., from commercial online experiment platforms, e.g., Mechanical Turk, or Microworkers) or amusement (i.e., from social media platform, e.g., Facebook) [13].

Although challenging, these differences can be overcome through careful design and by embedding reliability checks within the experiments. Previous works [17, 54] summarized the recommendations and best practices for online QoE assessment of multimedia, providing extensive evidence that by paying attention to the differences listed above in the experimental design, results as reliable as those collected in the lab can be obtained in online settings [14].

2.4 Facebook as a Research Tool for Online Experiment

A number of studies have shown that social media profiles, such as Facebook, convey fairly accurate impressions of the profile owners [21, 55, 79–81]. For example, Facebook profiles are shown to reflect personality of the profile owners in real-life [23], e.g., there is a positive correlation between a user's number of Facebook friends and a user's extraversion [20]. In addition, users typically use Facebook to maintain offline relationships with their friends [21] by sharing photos/videos or posting comments [56]. Videos spread much faster on social media (e.g., Facebook, Twitter) than on a video/image sharing network (e.g., YouTube) [57]. In short, Facebook provides a compelling source of measurable user information [58].

A growing number of researchers, especially in psychology and social science, have begun to collect user information via Facebook for research purposes [17]. User information from Facebook includes self-reported information (e.g., age, gender, current workplace, etc.), digital traces of behavior (e.g., likes, posts, photo uploaded, etc.) and information contributed by others (e.g., photo tagged or comments on a user's wall) [17]. Many studies have tried to estimate individual characteristics (e.g., personality, life satisfaction) based on user information collected from Facebook. For instance, user's personality traits are found to be significantly correlated with semantic features of status updates [59], Facebook profile [23], number of friends [60], number of photos

²<http://projectimplicit.net/>.

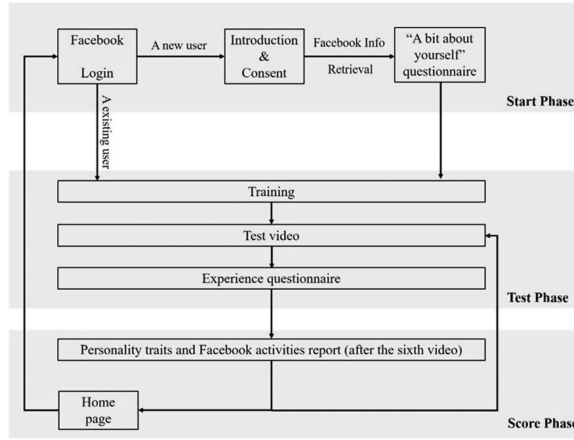


Fig. 1. The overall design of YouQ.

uploaded [61], profile photos [83] and so on. Facebook ‘likes’ can be used to draw predictive profiles of the Facebook user’s race, sexual orientation, religion, IQ, and the like [62]. Users’ personal interest can be estimated by analyzing the self-statement in their profiles as well as their previous activities e.g., videos watched or liked [63, 64]. Further, there has been work studying the difference between recruiting users from Facebook, paid Microworkers crowdsourcing platforms, and traditional controlled lab environments [65].

Of course, due to privacy concerns, users have the right to limit access to parts of their Facebook information [17, 21]. Thus, a well-designed Facebook application is needed to engage users and make them willing to share their Facebook information with researchers. MyPersonality, developed by [66], is a successful Facebook application to collect user information through an online personality survey. The application attracted more than 10 million users, and about 30% of its users were willing to share their Facebook information with the authors. The above examples thus demonstrate the potential of Facebook applications as a research tool to run subjective experiments by simultaneously collecting users’ information.

3 YOUQ: THE STRUCTURE AND DESIGN

In this article, we developed a Facebook application (which obtained approval from Facebook), YouQ, for the collection of individual video QoE data. YouQ has multiple advantages (1) it can reach users with diverse demographics in real-life viewing environments; (2) it can collect user information (e.g., age, gender, video watch history, uploaded pictures) automatically from Facebook.

In addition, the app allows checking for user reliability at every stage of the experiment. The app is open-sourced³ for research. Researchers are free to edit the set of stimuli and questionnaires based as need be.

3.1 Overall Description

YouQ organizes the entire process of online subjective testing in three main phases: (1) *Start Phase*, (2) *Test Phase*, and (3) *Score Phase* as shown in Figure 1. At the beginning of the *Start phase*, the user reaches the app page, and can log in from his/her Facebook account to access the app. New users (i.e., users who have not used the app before) are shown a short introduction and are asked consent to let YouQ retrieve the information he/she has shared on Facebook. Returning users can read

³The code is available at: https://drive.google.com/open?id=0B5_9Jyayi1XDWU5IV3N5UkIxa0k.

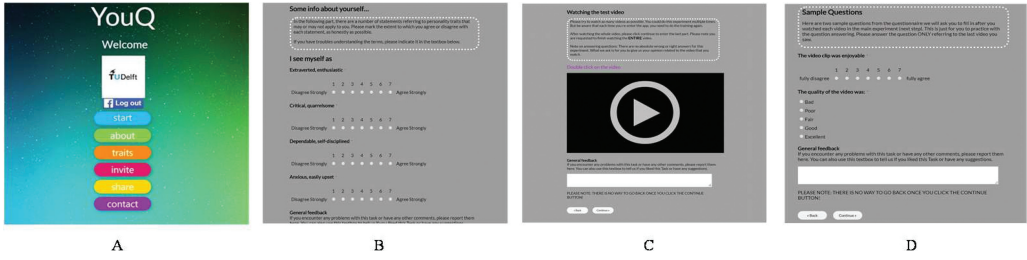


Fig. 2. The screenshots of YouQ: (A) represents the welcome page of YouQ; (B) represents partially the questionnaire collecting personal information; (C) represents the test page of YouQ; (D) represents a sample questionnaire regarding individual QoE.

this information again anytime by simply clicking the “About” button on the home page shown in Figure 2. A questionnaire is then presented to collect personal information: in our case, demographics, personal interest on different video genres, and Big-Five personality. By doing so, we intended to check whether such self-reported information matches the information automatically collected from Facebook. In parallel all the requested user Facebook information is retrieved and stored in a database (see Section 3.2 for a detailed explanation).

In the *Test phase*, users perform the actual QoE experiment. In the case of YouQ, it consists of video QoE rating. When the user starts this phase from the home page, and s/he is already registered in the database, s/he is asked to perform a short training before the actual test video takes place. There, the user has to watch two sample videos at two different distortion levels (to get familiar with the perceptual quality range spanned by the test stimuli) and perform QoE evaluation as it would be done during the test. Then, the actual experiment takes place and the user watches a video with a given bitrate/quality. Next, s/he is asked to rate the viewing experience, in our case through a questionnaire (explained in detail in Section 3.4). In order to ensure the experiment will not be affected by some unwanted impairments (e.g., buffering), it is necessary to pre-cache all data required during the experiment [46]. In other words, a video will not be played if it does not fully download in the experiment. Once the user finished the evaluation, the data, i.e., the name of the video watched and the questionnaire answers are stored and the user is redirected to the next phase.

In the *Score phase*, the user first gets the summary of his/her personality traits and is then asked how s/he wants to proceed. Note that the user is not required to evaluate six (or more) videos. After each evaluation and score phase, the user can choose either to evaluate more videos or to quit the test and application. If the user chooses to score other videos, s/he will be redirected to the test phase, where she/he will be scoring videos directly, without going through the training phase again.

3.2 User Information Collected in YouQ

Users volunteer to share their Facebook information as part of YouQ application (note that the user retains the right to decide what to share and what not to). YouQ asks for eight types of Facebook information. Requested data are the information in one’s Facebook profile, including gender, age, as well as the user’s list of friends. In addition, YouQ gets access to the list of all pages and Open Graph objects that a user has liked, the photos and videos a user has uploaded and been tagged in, and the posts on a user’s timeline, including his/her own and others posts. Note that YouQ can access this information once a user agrees to give YouQ the access. Such personal information is shown to capture many individual differences (i.e., personality [60, 61, 80, 81], demographics [67], health [79], socioeconomic status [68] etc.) and consequently might be predictive of individual QoE

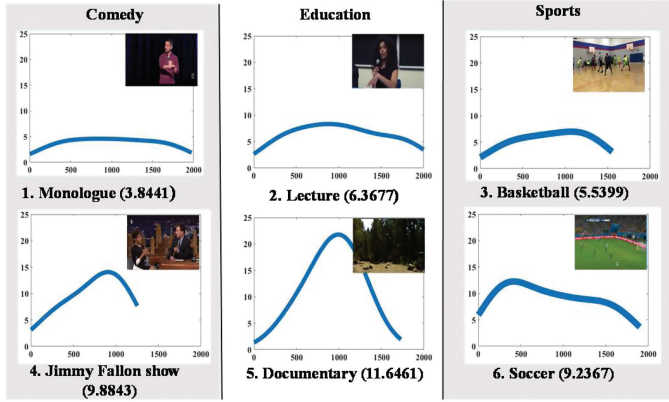


Fig. 3. The video content with its arousal curve used in this experiment. Here, Y-axis represents the arousal value (the intensity of emotion the video represents) whereas the X-axis represents the frame number. The video 1, 2, 3 represent videos with low excitement level whereas video 4, 5, 6 represent videos with high excitement level. The number between brackets represents the mean value of the arousal curve.

as well. For example, the total number of Facebook friends, uploaded photos/videos, likes, posts are proven to be significantly correlated with personality [60, 61], and therefore, are collected by YouQ.

3.3 Reliability Control Mechanisms

We introduce 5 types of reliability checks – before, during and after the experiment as suggested in [54]:

1. When a user logs in with his/her Facebook account and agrees to share profile information, YouQ collects his/her age/gender/ nationality. It then checks whether such information matches the information s/he has filled in the initial questionnaire (from the start phase). Those who do not provide consistent information are not considered in the analysis.
2. In the initial questionnaire, YouQ includes one reliability question to check whether the user can understand English. The user will be asked to indicate the resolution of their display. For those who don't know the answer, it is also fine to just type in "DK". Those who do not provide a resolution value or "DK", demonstrating they either don't have sufficient English proficiency or didn't pay attention to the question, are not considered in the analysis.
3. Consuming media with other tasks is an observed behavior in many people [69] e.g., the study has shown that around 75% users are multitasking while watching television [69]. To make sure users from the online experiment pay attention to the video and are able to understand the video, we included a very simple content question (a multiple choice, e.g., a man telling a joke for Figure 3, video 1) after each video, checking whether participants paid attention to the video they just watched. Data from participants who do not answer this question correctly is not considered in the analysis.
4. YouQ tracks the time a user is staying on each page by collecting the timestamps. If a user stays on a page far more than it normally takes (two times longer/shorter than the reference time), the user's response will be discarded in the analysis, as it may indicate that the user has been multi-tasking while evaluating the stimuli [54]. Note that the reference time varies

depending on the task and the length of the test video. In this article, the reference time is set based on a pilot experiment which involved five users. The average time spent on the task is defined as the reference time.

5. The questionnaires are answered on a Likert scale built out of radio buttons. All questions must be answered, otherwise the user cannot continue.

3.4 Questionnaires Used in YouQ

YouQ has been designed to collect user information automatically (as explained in Section 3.1) together with subjective QoE ratings. Specifically, YouQ measures the user's viewing experience in terms of perceived video quality and enjoyment, as it was proven in previous research that these two aspects are core to characterizing Quality of Experience [70]. For perceived video quality, YouQ asks the user to indicate the perceived overall quality of a video he/she just watched on an ACR (i.e., Absolute Category Rating) scale [6]. Besides perceived video quality, the measurement of QoE should also be complemented by a measurement of the level of enjoyment of the experience, which reflects how much happiness or fun a user gets from the video [70]. As a result, YouQ asks users to indicate their level of enjoyment for the video they just watched on a 7-point Likert scale. The app is also configurable to add further measurements of the viewing experience evaluation, such as endurance, satisfaction, or involvement [11].

In addition, for this pilot phase of the app, we introduced a short introductory questionnaire (see "start phase" in Section 3.1), where we collected self-reports of user data to cross-check with the Facebook data collected automatically. The introductory questionnaire asks participants to indicate their age, gender, nationality, and levels of interest in different video genres (on a 7-point Likert scale). The "a bit about yourself" personality questionnaire, mainly adapted from [71], is then presented, consisting of 10 items to be scored on a 7-point Likert scale. The aggregated questionnaire results provide a personality profile along the "Big-5" traits [71], including Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism.

4 EXPERIMENTAL SETUP

The main goal of this article is to explore the suitability of Facebook to function as an experimental platform for video individual QoE research. More concretely, the goal is to create a Facebook experimental environment that is able to mimic users' viewing experience in real life, reflecting users' natural behavior. As a first step, we intended to verify whether the App we designed would provide reliable results when launched "in the wild", i.e., openly on Facebook without supervision and controlled settings, as compared to a classic laboratory setting. As a second step, we wanted to check that the results obtained were also in line with the existing literature on individual QoE [11, 12, 84–86]. To this end, we designed an experiment with a setup similar to that used for two recent studies in the lab environment [11, 12], investigating the effect of personality and demographics on perceived quality and enjoyment of videos of varying bitrate, genre, and affective content [11, 12].

We performed this experiment by using YouQ in two different sessions. In the first one, participants were invited to use the app in a controlled laboratory setting, with an experimenter supporting them throughout the task. In the second session, the app was launched on Facebook and heavily advertised throughout the personal networks of the authors. In the following, we describe the experimental setup for both sessions in more detail.

4.1 Stimuli

We used six video contents from three genres (two contents per genre) in this experiment (shown in Figure 3), chosen according to the indications of [11]. All video contents are originally from

YouTube and in English. These videos were chosen to match the content associated with user's consumption of videos on a day-to-day basis [73]. In the traditional video quality evaluation, users usually are asked to watch short sample (i.e., around 10 seconds) video with different visual impairments in it. Such short videos usually fail to reach user engagement and attention closer to real-life video viewing [72]. Authors from [72] suggested at least 1-minute long videos to be able to engage users. In addition, users from an online experiment can easily leave the experiment if it is too long. Authors from [14] suggested that online experiment should not be longer than 10 minutes to keep users motivated. Therefore, the duration of all test videos was limited to between 1 and 2 minutes to make sure users can engage in the video while being able to finish the whole experiment in 10 minutes.

Videos were varied in terms of their affective content. The affective content of a video (i.e., the intensity of emotion that is expected to arouse in the user while watching that video [66, 87]) is suggested to influence individual viewing experience, and has been previously investigated within individual QoE in [11] and [87]. For this reason, in our experiment, we selected per each genre two content types, at high and low excitement level, respectively. To determine the excitement level of a video, we took the mean value of the arousal curve (as shown in Figure 3) throughout the whole video, calculated according to [73]. During our lab experiment, we asked users to rate the affective response for each test video on a 7-point Likert scale (i.e., reflecting how exciting the video that the user just watched was). After collecting the data, a linear mix model was used to validate the manipulation of affective content. Here, the affective ratings were considered as dependent variable, whereas the excitement level, bitrate, and genre were considered as independent variables. The result shows that the test videos with high excitement levels were rated significantly higher than those with low excitement level ($F = 60.284, p < .000$).

All videos were encoded in H.264/AVC, 30 frames per second. To represent different video quality conditions, we encoded each video at two bitrate levels, i.e., 2000kbps and 500kbps, according to what was done in [11]. As a result, a total number of 12 videos are used in this study. The audio configuration of videos was kept constant, in the AAC format with a bit rate of 112kps, to avoid any effect of audio quality on overall QoE. The video resolution was set to 896×504 pixels, which is the default size of a YouTube video player, to accommodate for different screen resolutions.

Finally, two 10s video samples are used for training. One sample is encoded at 500kbps (representing lowest quality) while the other one is encoded at 2000kbps (representing highest quality). By doing so, we intend to make the users acquainted with the range of perceptual quality of the videos we have. The media configuration of these samples is the same as for the test videos (30fps, 896×504 , H.264 AVC). All videos used in this study were encoded and stored on our server.

4.2 Procedure

We performed this experiment via YouQ both in the lab and in the wild in order to compare the results.

Lab Experiment. In the lab setting, users sat in front of a 27-inch display. The environmental setup, including the lighting settings, was strictly according to the ITU recommendations [6]. Users were greeted and explained the purpose of the experiment. They were then asked to log into YouQ, which was displayed on a Google Chrome browser, with their Facebook account, and to give us the permission to retrieve their information from Facebook. They then reached a welcome page with a description of the experiment followed by a short consent form. After giving consent, the user started the experimental session, going through the three phases described in Section 3. First, the user filled in the personal information requested by the starting questionnaire described in Section 3.4. Then, s/he performed the training session to understand the task and visualize the different quality levels presented in the experiment. The test phase would then start, with the

Table 1. Demography of the Users in Online and Lab Experiment

	online	lab
No. of users	136	20
Age range	Min: 15, Max: 70, Mean: 31	Min: 20, Max: 32, Mean: 27.8
Gender	66 Female, 70 Male	5 Female, 15 Male
resident	17 countries	Netherlands

user watching all 12 test videos after the training, presented in a random order to avoid fatigue and memory effects. Note that in the lab experiment all users watched all 12 videos in a within-subjects setting, given that there were no constraints on the brevity of the task. The experiment took about 40 minutes to finish, and users got their respective personality traits after watching the first 5 videos.

In-the-Wild Experiment. For the in-the-wild session of the experiment, YouQ was openly published on Facebook, and users were recruited indirectly through advertisement. Also, in this case, when logging into the app for the first time, the user has to give consent for us to access his/her data and fill in the questionnaire on personal data. After that, the test phase (see Section 3) starts, and users complete the evaluation of at least one video in a given bitrate. The test video is randomly selected from the video pool. After finishing the first test video (and receiving the first bit of information on their personality traits), the users could choose to quit the application or to continue and evaluate another test video, until they see at most one version of each of the six video contents. As a result, the within-subject design adopted in the lab was mostly dismissed in this phase, mainly to ensure that the task was kept short and engaging. It takes around 10 minutes to finish the questionnaire plus the first test video.

5 RESULTS

5.1 YouQ in the Wild and in the Lab

Table 1 presents an overview of the participation in the two experimental sessions. In the lab experiment, we had 20 users (5 Female and 15 Male), all of them students from Delft University of Technology, aging from 20 to 32 with the mean age 27.8 years. Two hundred and twenty-three (223) users joined in the online experiment, 89 users failed to pass the reliability checks (e.g., inconsistency between self-report information and automatic Facebook information retrieval). As a result, 136 users (66 Female, 70 Male) were recruited in the online experiment, aging from 15 to 70 with the mean age 31, and coming from 17 countries (e.g., USA, Canada, Netherlands, and so on). Each user rated an average 1.55 videos and, on average, each test video received more than 15 ratings.

In order to check whether the online experiment can collect reliable results as compared to the lab experiment, we built a linear mixed model where bitrate level, excitement level, genre and experiment condition (in-the-wild or lab) were considered as fixed factors, and user was considered as one random factor. Model parameters were determined with the restricted maximum likelihood method (REML). The perceived quality ratings (here, PQ indicates the perceived overall quality of a user with a video with the Absolute Category Rating [6]) and enjoyment were used as dependent variables. The rationale behind this type of analysis is that, if the in-the-wild experiment was not reliable, we would obtain different effects of the fixed factors on enjoyment and perceived quality and the fixed factor experiment condition would have a significant effect on the dependent variables. It should also be noted that, although in previous comparisons between lab- and crowdsourcing-based experiments [19, 54] result reliability was checked by, e.g., computing the

Table 2. The Linear Mixed Models for Enjoyment and PQ on Online and Lab Experiment. PQ Indicates the Perceived Overall Quality of a User with a Video with the Absolute Category Rating (from Bad to Excellent)

Source	Enjoyment		PQ	
	F	Sig. (p)	F	Sig. (p)
Experiment Condition	.058	.810	2.289	.131
Intercept	2236.96	.000	3634.08	.000
Bitrate	0.082	.774	9.298	.000
Excitement	43.31	.000	.032	.859
Genre	0.099	.906	.225	.799

*Sig. stands for significance.

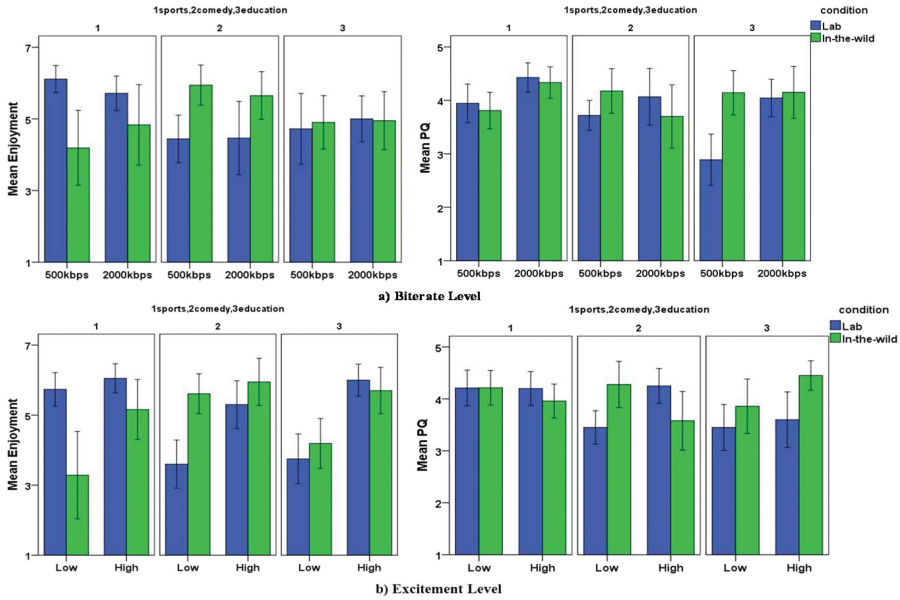


Fig. 4. Enjoyment and perceptual quality ratings obtained in the lab (blue columns) and in-the-wild (green columns). In Figure 4(a), Y-axis represents the mean value of enjoyment and PQ, whereas the X-axis represents two bitrate levels. In Figure 4(b), Y-axis represents the mean value of enjoyment and PQ whereas the X-axis represents two excitement levels.

correlation between the MOS obtained in lab and crowdsourcing, for our purposes (i.e., measuring individual QoE), this has little meaning, as we are especially interested in the individual differences. The mixed model we use allows us instead to still check that the main effects are consistent throughout experimental conditions (i.e., between lab and in-the-wild), yet accounting for individual differences (modeled through the random factor user).

As shown in Table 2, the excitement level ($F = 43.31$, $p < .000$) has a significant impact on enjoyment whereas the bitrate level ($F = 9.298$, $p < .000$) has a significant impact on PQ. There is no significant difference between the evaluations of enjoyment and PQ performed in two experimental conditions (i.e., in the lab and in the wild). For enjoyment, the larger confidence intervals were found in the wild condition, both per bitrate level and per excitement level (as shown in Figure 4), this might due to the fact that enjoyment is highly related to the user's interest [11] and personal

Table 3. A Summary of the Experimental Settings of the CP-QAE-I and i_QoE

		CP-QAE-I	i_QoE
Users	Number	114	59
	Nationalities	18	13
	Gender	81Male/33Female	27Female/32Male
Outcomes		Enjoyment, Perceived Quality	Enjoyment, Perceived Quality
Input Questionnaires		BFI-10, Hofstede Culture Questionnaire, Gender, Nationality	BFI-10, Interest, Gender
Stimuli	Videos	12	6
	Number of Videos	144 clips with varying quality conditions	12 clips with two bitrate levels

taste. The users from the online experiment might be more divergent in terms of personal interest as compared to those from the lab experiment, which was mostly from the Computer Science department in a technical university. On the other hand, similar confidence intervals were found in both conditions on perceived quality suggesting all users have similar interpretation of video quality regardless which condition the user in.

5.2 A Systematic Comparison between Two Recent Studies and YouQ on Individual QoE

The next step in verifying the suitability for Facebook to function as a platform for the collection of individual QoE rating is the comparison between the results from our online experiment with two previous independent studies (i.e., i_QoE [4] and CP-QAE-I [12]).

5.2.1 Two Systematic Studies on Individual QoE. To the best of our knowledge, only two studies, i_QoE [11] and CP-QAE-I [12] have investigated systematically the influence of user factors on individual Quality of Experience. These studies have targeted not only perceptual quality but also and explicitly enjoyment for the experience, to obtain a more holistic view on what Multimedia (in this case, video) experiences, are and how user factors impact them. Table 3 summarizes the experimental settings of the two studies. i_QoE found a weak correlation between perceived quality and enjoyment. Videos delivered at a low bitrate were reported as having significantly lower perceived quality than videos with maximum bitrate (about 2000Kbps). A corresponding drop in enjoyment was, however, not recorded. Looking at user factors, an influence of cultural background was found: users from Asia (e.g., China, India) tended to rate their enjoyment higher than those from western countries (e.g., UK, the Netherlands). With respect to demographics, gender was shown to affect the user's level of involvement. In particular, males were easier involved with the video content than females in this study, and this trend was independent of the video genre. Concerning affective aspects, the level of interest that the user had in the genre of the video s/he was about to watch was found to be positively correlated with enjoyment ratings. Finally, a significant correlation between enjoyment and one of the big 5 personality traits, i.e., extraversion, was observed.

CP-QAE-I also showed that lower system parameters (i.e., frame rate, bitrate, and resolution) lead to lower perception of video quality, but not necessarily lower enjoyment. Lower frame rate had a negative effect on user's level of enjoyment whereas lower bitrate and resolution did not. Especially for content videos with objectionable content (graphic murders etc.), no significant correlation between quality and enjoyment was found, which may be related to the level of interest

Table 4. Baseline Model for Enjoyment on YouQ, i_QoE and CP-QAE-I

Baseline	Enjoyment	YouQ			i_QoE			CP-QAE-I		
	Source	De. df	F	Sig.	De. df	F	Sig.	De. df	F	Sig.
Baseline	Intercept	179.5	2071.9	.000	319	2960	.000	1183	7157	.000
	Bitrate	177.31	.920	.433	304	1.564	.212	1163	.463	.496
	Content	62.665	14.277	.000	107	17.56	.000	187	39.21	.000
Optimistic	Intercept	159.89	2028.8	.000	129.5	2493	.000	209.1	3066	.000
	Bitrate	137.59	.829	.480	111.9	1.388	.241	209.3	0.038	.846
	Content	58.9	14.398	.000	113.9	19.92	.000	184.2	47.811	.000

Table 5. Baseline Model for Perceived Quality on YouQ, i_QoE and CP-QAE-I

Baseline	PQ	YouQ			i_QoE			CP-QAE-I		
	Source	De. df	F	Sig.	De. df	F	Sig.	De. df	F	Sig.
Baseline	Intercept	202	3822.9	.000	259	8394	.000	1159	8559	.000
	Bitrate	102	6.748	.000	237	52	.000	1186	10.78	.001
	Content	202	3.936	.002	133	11	.000	209	9.021	.000
Optimistic	Intercept	163.19	3833.4	.000	140	5943	.000	217.85	4087	.000
	Bitrate	148.70	6.991	.000	96.769	29.49	.000	213.25	4.698	.031
	Content	71.533	4.883	.001	105.55	15.83	.000	179.06	10.98	.000

*De. df. stands for Denominator degrees of freedom, Sig. stands for significance.

users had in them (although in this study interest was not measured). Enjoyment was found to be significantly correlated with a subset of positive affect traits [12], i.e., joy, satisfaction, and interest (note that, here, interest refers to user's response to the video they just watched, which is different from the definition of "interest" used in i_QoE study). Finally, neurotic users tend to enjoy videos with negative emotions (e.g., sad, fearful, guilty, or ashamed) more.

5.2.2 Comparison. We compare these two studies with the results from our online experiment by using an incremental approach. We use linear mixed-effects models with repeated measures, with model parameters determined with REML, following the analysis method used in our previous study [12]. We only consider common user factors that are available for all three studies (i.e., gender and five personality traits) as input for the analysis. Perceived Quality and Enjoyment are treated as dependent variables separately. We first analyze the outcomes of three studies separately, and then we outline similarities and differences to understand how user factors play a role in individual QoE.

We first build baseline models which only consider common system factors of three studies, i.e., bitrate and video content (considered as two fixed factors). The results for Enjoyment and Perceived Quality are shown in Tables 4 and 5, respectively. It can be seen that both bitrate and video content have a significant effect on the Perceived Quality in all three studies. With regard to Enjoyment, video content has shown a significant impact whereas bitrate has not for all three studies.

Secondly, we build optimistic models to provide an estimate of the proportion of residual variance which could be reasonably attributed to user factors [12]. The optimistic models consider each user as a random factor, estimating intercept and slope of the linear model per each user separately. This allows the model to explain more variance by taking into account individual differences. The optimistic models are then compared with the baseline models using paired t-tests

Table 6. Paired T-test Comparing Optimistic Models and Baseline Models for Both Datasets on MSR

		Mean	Std.	Std.	95% CI		t	df	Sig.
			Deviation	Error Mean	Lower	Upper			
YouQ	Enjoyment	1.649	2.598	.179	1.297	2.00	9.223	210	.000
	PQ	.471	.714	.049	.374	.568	9.583	210	.000
i_QoE	Enjoyment	8.940	20.82	1.106	6.765	11.117	8.081	353	.000
	PQ	.183	.511	.027	.129	.236	6.722	353	.000
CP-QAE-I	Enjoyment	.419	1.130	.032	.357	.483	13.038	1231	.000
	PQ	.362	1.015	.029	.305	.419	12.523	1231	.000

*Std. stands for standard.

on the Mean Squared Residuals (MSR), shown in Table 6. A significant increase in the amount of variance explained by the optimistic model with respect to the baseline one can then be attributed to the presence of the user random factor, and indicates that by incorporating knowledge on individual differences, more accurate models can be devised.

YouQ. For Enjoyment, the MSR in the baseline model is 2.42. The optimistic model reduces the MSR to 0.77 ($p < .000$) implying that 68% of the predicted variance is attributable to individual differences. For Perceived Quality, the MSR is reduced from 0.824 to 0.626 ($p < .000$) from the baseline to optimistic model, indicating that an additional 24% of the variance can be explained by the optimistic model, and hence by user factors

i_QoE. For Enjoyment, the MSR in the baseline model is 36.945. The optimistic model reduces the MSR to 28.004 ($p < .000$) with an increase of 15.7% of the overall variance predicted. For Perceived Quality, the MSR is reduced from 0.759199 to 0.614903 ($p < .000$) from the baseline to optimistic model, representing a predicted variance of 15.7% attributable to users.

CP-QAE-I. For Enjoyment, the MSR in the baseline model is 1.3885. The optimistic model reduces the MSR to 0.9686 ($p < .000$) with an increase 24.9% of the overall variance predicted. For Perceived Quality, the MSR is reduced from 1.3547 to 0.9926 ($p < .000$) from the baseline to optimistic model, hence the user factor allows an increase of 22.1% in the variance explained by the model.

The results above illustrate the importance of user factors since a large proportion of variance can be explained by considering users as a “random effect”, especially on the results of YouQ. This suggests that the use of online testing may reveal even more about individual differences in QoE, yet preserving reliable results with respect to the effect of fixed factors on QoE ratings.

Modeling users as random factors is useful to quantify the impact of user factors on QoE, but has limited applicability when the models are to be deployed in real time to steer individual QoE management, as it is impossible to know *a priori* which intercept and slope will correspond to each user. In that sense, it is more convenient to identify individual traits that characterize the user and impact individual QoE, to be used as fixed factors and co-variates in fully defined models. To this end, the next step in our analysis is to identify the most informative user factors in predicting the experience of Enjoyment and/or Perceived Quality. Note that we only consider the common factors of three studies (YouQ, i_QoE, CP-QAE-I) as covariates with direct effects (i.e., five personality traits and gender). Again, bitrate and video content are considered as two fixed factors. The rest of the setup is kept the same as for the baseline models.

The results of the extended models are shown in Tables 7 and 8. All significant effects are marked in red. For YouQ, two personality traits, i.e., an extroverted ($F = 7.6$, $p = 0.007$) and open ($F = 4.134$, $p = 0.044$) user significantly enjoyed the content more. On the other hand, no significance is found to influence Perceived Quality. For i_QoE, an extroverted user ($F = 11.933$, $p = 0.001$) is also found to significantly enjoyed the content more. A user who has a more agreeable personality tends to

Table 7. Extended Model for Perceived Quality on YouQ, i_QoE, and CP-QAE-I

Source	YouQ		i_QoE		CP-QAE-I	
	F	Sig.	F	Sig.	F	Sig.
Intercept	67.436	.000	154.632	.000	67.315	.000
Bitrate	7.038	.000	51.137	.000	11.932	.001
Content	4.593	.001	11.982	.000	9.045	.000
Gender	1.159	.284	3.738	.054	.568	.451
Extraversion	.000	.994	.035	.852	.001	.976
Agreeableness	.009	.925	7.190	.008	3.069	.080
Conscientiousness	.169	.682	.467	.495	5.382	.021
Neuroticism	.122	.728	2.504	.115	5.934	.015
Openness	.303	.583	1.138	.287	.788	.375

Table 8. Extended Model for Enjoyment on YouQ, i_QoE, and CP-QAE-I

Source	YouQ		i_QoE		CP-QAE-I	
	F	Sig.	F	Sig.	F	Sig.
Intercept	8.599	.004	18.330	.000	62.918	.000
Bitrate	1.170	.324	1.739	.188	.433	.510
Content	12.683	0	18.894	.000	38.718	.000
Gender	.174	.677	.077	.781	9.425	.002
Extraversion	7.600	.007	11.933	.001	.934	.334
Agreeableness	.873	.352	.135	.714	3.990	.046
Conscientiousness	.770	.382	2.507	.114	2.808	.094
Neuroticism	.269	.605	.000	.984	.000	.984
Openness	4.134	.044	.211	.646	3.543	.060

rate the perceived quality significantly higher ($F = 7.190$, $p = 0.008$). For CP-QAE-I, female users tend to ($F = 9.425$, $p = 0.002$) enjoy the video more as compared to male users. Similar results can be found for agreeable users ($F = 3.990$, $p = 0.046$) w.r.t. their ratings on enjoyment. A user who is conscientious ($F = 5.382$, $p = 0.021$) rates perceived quality of a video significantly more.

5.2.1 Discussion. The previous analysis is, to the best of our knowledge, the first meta-analysis across the existing systematic studies of the influence of user factors on individual Quality of Experience. From the results above, we find three major points:

1. *The investigated system factors* do not influence user's level of Enjoyment, though they have a clear negative effect on user's Perceived Quality, as expected from literature. Lower bitrate corresponds to lower Perceived Quality. On the other hand, video bitrate level has shown no impact on Enjoyment, implying users may still be satisfied with the experience, even if bit rate decreases a bit. As Enjoyment is strongly influenced by individual differences, user factors should play a more prominent role in future video delivery optimization mechanisms.
2. *User factors* represent a large proportion of variance for both Enjoyment and Perceived Quality. The results of YouQ from our online experiment indicate that user factors play a major role in determining individual experience. However, no single user factor is found to have a significant impact on either Enjoyment or Perceived Quality across the three studies. This may be due to the fact that the three studies have only a limited number of

Table 9. Correlations between Facebook Information and Enjoyment and PQ on YouQ

Source	Enjoyment		PQ	
	Coeff.	Sig.	Coeff.	Sig.
user_Friends	.098	.156	.092	.186
user_PhotosUpload	.054	.437	.040	.566
user_PhotosTagged	-.026	.711	-.088	.204
user_Status	.142	.043	.111	.114
user_Likes	.146	.035	.086	.214

user factors in common, and that those are not the most informative ones. For example, i-QoE found a strong impact of interest on Enjoyment, but this factor was not considered in CP-QAE-I, and thus not included in the analysis. In fact, interest is significantly correlated to Enjoyment for YouQ ($r = 0.295$, $p = 0.01$) suggesting users who interested in the certain genre (e.g., comedy) enjoyed the video of that genre more. The result implies that more user factors should be collected/considered in the future QoE studies, and also that a large QoE dataset with user factors is badly needed.

3. *Content matters.* Video content itself has been found to have a significant influence on individual QoE in all studies. Except for the above-mentioned relationship between genre and amount of movement in the video (see Section 2.1.1) which may impact perceptual quality and enjoyment as a consequence, the differences in enjoyment between different video content may relate to user factors (e.g., personality). For instance, users with a certain type of personality (e.g., neurotic) tend to enjoy a certain type of video (e.g., sad or fearful).

5.3 The Impact of Users' Facebook Profile on Both Enjoyment and Perceived Quality

As introduced in Section 3.2, user information was collected automatically from the Facebook profiles of users of YouQ during the Start phase in our experiment. This data could be used to further improve our individual QoE models, having the advantage that it can be collected automatically once the user has given consent. To show the usefulness of this Facebook information, as a first attempt, we performed a spearman rank order correlation analysis between the Facebook information and three individual QoE measures. Here, only the data from the online setting was considered. Since users tend to like the content they enjoy, ratings on Enjoyment are found to have a positive significant correlation with the number of likes users gave (Table 9). However, there is no significant correlation between PQ and the Facebook information.

Additionally, all types of Facebook information were normalized and put into a linear model. Here, PQ and Enjoyment were considered as three dependent variables separately, whereas bitrate and video content were considered as two fixed factors; age, gender, total numbers of photo uploaded, photo tagged, likes, friends and posts, were considered as covariates. The results are shown in Table 10: the number of user likes was found to have significant impact on enjoyment ($F = 4.172$, $p = 0.043$), no significance is found on PQ. These results indicate that Facebook information may have an impact on individual viewing experience (i.e., enjoyment) and should be further investigated. However, it is important to remark that these results are preliminary and more data needs to be collected to further substantiate them. The use of non-linear models for the prediction of Enjoyment, PQ may be beneficial to uncover complex interactions among the different types of user information.

Table 10. Linear Model for Enjoyment and PQ on the Importance of Facebook Information

Source	Enjoyment		PQ	
	F	Sig.	F	Sig.
age	2.040	.156	.000	.983
gender	.127	.722	.275	.601
user_Friends	.845	.360	2.297	.132
user_PhotosUpload	.859	.357	.000	.982
user_PhotosTagged	2.167	.145	3.140	.079
user_Status	.200	.656	2.676	.104
user_Likes	4.172	.043	.033	.856

6 CONCLUSION

There is no such thing as an “average” user. The next generation of multimedia services have to be geared towards personalization and multimedia delivery needs to start taking individual differences and preferences into account. The associated costs of doing this, in terms of efforts and resources to conduct subjective tests, is, however, a challenge. From a modeling standpoint, it also implies the adoptions of methods which automatically sense user properties and incorporate them in existing quality metrics and models.

In this article, we have furthered the understanding of individual Quality of Experience by reporting on the results of a study involving Facebook-based subjective experiments. We designed YouQ, a Facebook application created to allow QoE evaluations of video, which can be configured to collect both self-reports of user information as well as personal data enclosed in the user’s Facebook profile. We have shown three important outcomes. First, when launched on Facebook, YouQ allows us to collect data as reliably as in classic lab experiments. In addition, it allows us to reach out to a much more diverse set of users (with a higher gender balance and a greater diversity in cultural backgrounds) with comparatively lower effort (for instance, the experimenter does not need to be present during the experiments), and in a timespan comparable to that of classic lab experiments.

Second, through the first meta-analysis of studies on individual QoE, we showed that individual QoE evaluations collected through YouQ are consistent with those reported in the literature [11, 12]; moreover, they also capture additional individual differences, with a large part of the data variance explained by modeling the user as a random factor. This may, in part, be explained by the fact that every user who accessed the app performed the experiment in a different environment and different device. On the other hand, crowdsourcing studies have shown that the impact of different viewing and environmental settings on QoE ratings may be minimal. Hence, we can hypothesize that the greater diversity in user responses is also due to the greater diversity in user background and demographics, allowed by the use of the Facebook platform. We are still actively recruiting participants and expect to have a follow-up study on a much larger cohort. The reason that we want YouQ to go public is for more people in the field to see the value of a Facebook-based experiment and build a great QoE test platform together with us.

Finally, we showed that the personal user information that YouQ automatically acquires from the user profiles can be useful in modeling individual QoE more accurately. For instance, the number of likes users gave is significantly correlated to their Enjoyment. However, it should be noted that, given the importance of online privacy and data protection, Facebook has currently restricted privacy regulations for its apps and it is rather difficult to get approval to collect user information

via Facebook apps, as users have the rights to refuse giving their information while using our app. Therefore, more fun and gamification elements should be added in YouQ to attract an even greater number users, concomitantly ensuring they are willing to share their Facebook information.

Our work raises the potential for some worthwhile future endeavors. Although clearly pointing out the importance of investigating individual differences in QoE, our results have also highlighted that the community is still far from having a clear picture of which user factors are relevant in QoE optimization, and to what extent. As a first priority, the community should aim at shedding some light on this issue. An inventory of potential factors should be created, and their influence systematically investigated across systems and services through empirical research and subjective studies.

Assessing the influence of these user factors on QoE will require extensive empirical research and involve a large number of users in order to have a diverse enough (in terms of individual differences in the factor under observation) sample population. In this sense, traditional lab-based subjective experiments may be limiting. Besides the well-known limitations in terms of ecological validity [76], lab-based subjective experiments are often designed to minimize, rather than embrace, individual differences. Our study proved that Facebook may be a valuable tool towards this end.

Of course, our ultimate goal is to achieve user-centered multimedia delivery optimization and adaptation. For accomplishing this goal, user information needs to be obtained and the user modeling community distinguishes between explicit and implicit methods for inferring user characteristics [13]. Explicit methods require the user to provide information directly, e.g., via web forms or questionnaires; Netflix, for example, asks new users for an indication of their movie preferences with a star-based voting system. When it comes to online optimization, for example for HTTP base adaptive video streaming, explicit user data collection is inconvenient, especially when dynamic user factors are of interest. Implicit, or unobtrusive, methods would be preferable in this case and the study reported in this article has indeed shown that social media can be used as a powerful research tool for future QoE evaluation. Nonetheless, more research needs to be done to establish the right mix of methods to collect user information, striking a balance between unobtrusiveness as well as user privacy concerns. We trust that the community will pick up this challenge and, in so doing, go some way towards providing a truly personalized user multimedia experience.

REFERENCES

- [1] Cisco Visual Networking Index. 2013. The zettabyte era—trends and analysis. *Cisco white paper* (2013).
- [2] Conviva. 2015. *Experience report*. Retrieved from <http://www.conviva.com/conviva-viewer-experience-report/vxr-2015/>.
- [3] P. Le Callet, S. Möller, and A. Perkis. 2012. Qualinet White Paper on definitions of quality of experience. *ENQEMSS (COST Action IC 1003)* (2012).
- [4] Y. Zhu, A. Hanjalic, and J. A. Redi. 2016. QoE prediction for enriched assessment of individual video viewing experience. *ACM Multimedia*, Amsterdam.
- [5] J. A. Redi, Y. Zhu, H. de Ridder, and I. Heynderickx. 2015. How passive image viewers became active multimedia users. Springer.
- [6] ITU-R BT.500-11, Methodology for the subjective assessment of the quality of television pictures. *International Telecommunication Union, Geneva, Switzerland*, 4 (2002), 2.
- [7] W. Lin and C.-C. Jay Kuo. 2011. Perceptual visual quality metrics: A survey. *JVCI* 22, 4 (2011), 297–312.
- [8] R. Hu and P. Pu. 2011. Enhancing collaborative filtering systems with personality information. *ACM Recsys*.
- [9] W. Wu, L. Chen, and L. He. 2013. Using personality to adjust diversity in recommender systems. *ACM Hypertext*.
- [10] M. Swan. 2012. Health 2050: The realization of personalized medicine through crowdsourcing, the quantified self, and the participatory biocitizen. *J. Personalized Medicine* 2, 3 (2012), 93–118.
- [11] Y. Zhu, I. Heynderickx, and J. A. Redi. 2015. Understanding the role of social context and user factors in video quality of experience. *Computers in Human Behavior* 49 (2015), 412–426.

- [12] M. J. Scott, S. C. Guntuku, Y. Huan, W. Lin, and G. Ghinea. 2015. October. Modelling human factors in perceptual multimedia quality: On the role of personality and culture. *ACM Multimedia*.
- [13] S. Möller and A. Raake. 2014. Quality of experience - advanced concepts, applications and methods. Springer International Publishing Switzerland 2014. <https://doi.org/10.1007/978-3-319-02681-7>
- [14] T. Hoßfeld, C. Keimel, M. Hirth, B. Gardlo, J. Habigt, K. Diepold, and P. Tran-Gia. 2014. Best practices for QoE crowdtesting: QoE assessment with crowdsourcing. *IEEE TMM* 16, 2 (2014), 541–558.
- [15] S. D. Gosling and W. Mason. 2015. Internet research in psychology. *AR Psychology* 66 (2015), 877–902.
- [16] L. J. Skitka and E. G. Sargis. 2006. The internet as psychological laboratory. *AR Psychology* 57 (2006), 529–555.
- [17] M. Kosinski, S. C. Matz, S. D. Gosling, V. Popov, and D. Stillwell. 2015. Facebook as a research tool for the social sciences: Opportunities, challenges, ethical considerations, and practical guidelines. *American Psychologist* 70, 6 (2015), 543.
- [18] A. Doan, R. Ramakrishnan, and A. Y. Halevy. 2011. Crowdsourcing systems on the world-wide web. *Communications of the ACM* 54, 4 (2011), 86–96.
- [19] J. Redi, E. Siahaan, P. Korshunov, J. Habigt, and T. Hossfeld. 2015. When the crowd challenges the lab: lessons learnt from subjective studies on image aesthetic appeal. *CrowdMM* 2015.
- [20] K. Casler, L. Bickel, and E. Hackett. 2013. Separate but equal? A comparison of participants and data gathered via Amazon’s MTurk, social media, and face-to-face behavioral testing. *CHB* 29, 6 (2013), 2156–2160.
- [21] R. E. Wilson, S. D. Gosling, and L. T. Graham. 2012. A review of Facebook research in the social sciences. *Perspectives on Psychological Science* 7, 3 (2012), 203–220.
- [22] F. Delogu, M. Franetovic, and L. Shamir. 2015. Keep me posted!: Human and machine learning analysis of facebook updates. *Mobile MM Communications*, 2015.
- [23] M. D. Back, J. M. Stopfer, S. Vazire, S. Gaddis, S. C. Schmukle, B. Egloff, and S. D. Gosling. 2010. Facebook profiles reflect actual personality, not self-idealization. *Psychological Science* (2010).
- [24] L. Gou, M. X. Zhou, and H. Yang. 2014. KnowMe and shareme: Understanding automatically discovered personality traits from social media and user sharing preferences. *ACM CHI*, 2014.
- [25] N. Seitz. 2003. ITU-T QoS standards for IP-based networks. *IEEE Communications Magazine* (2003), 82–89.
- [26] P. Brooks and B. Hestnes. 2010. User measures of quality of experience: Why being objective and quantitative is important. *Network, IEEE* 24, 2 (2010), 8–13.
- [27] F. Dobrian, V. Sekar, A. Awan, I. Stoica, D. Joseph, A. Ganjam, J. Zhan, and H. Zhang. 2011. Understanding the impact of video quality on user engagement. *ACM Computer Communication Review* (2011), 362–373.
- [28] G. Ghinea and J. Thomas. 2005. Quality of perception: user quality of service in multimedia presentations. *IEEE Transactions on Multimedia* 7, 4 (2005), 786–789.
- [29] S. Ickin, K. Wac, M. Fiedler, L. Janowski, J.-H. Hong, and A. K. Dey. 2012. Factors influencing quality of experience of commonly used mobile applications. *IEEE Communications Magazine* 50, 4 (2012), 48–56.
- [30] A. Balachandran, V. Sekar, A. Akella, S. Seshan, I. Stoica, and H. Zhang. 2012. A quest for an internet video quality-of-experience metric. In *Proceedings of the 11th ACM Workshop on Hot Topics in Networks*. 2012.
- [31] Q. Huynh-Thu and M. Ghanbari. 2008. Temporal aspect of perceived quality in mobile video broadcasting. *IEEE Transactions on Broadcasting* 54, 3 (2008), 641–651.
- [32] J. H. Westerink and J. A. Roufs. 1989. Subjective image quality as a function of viewing distance, resolution, and picture size. *SMPTE Journal* 98, 2 (1989), 113–119.
- [33] N. Staelens, S. Moens, W. Van den Broeck, I. Marien, B. Vermeulen, P. Lambert, R. Van de Walle, and P. Demeester. 2010. Assessing quality of experience of IPTV and video on demand services in real-life environments. *IEEE Transactions on Broadcasting* 56, 4 (2010), 458–466.
- [34] K. Yamori and Y. Tanaka. 2004. Relation between willingness to pay and guaranteed minimum bandwidth in multiple-priority service. *Communications*, 2004.
- [35] C. Owsley, R. Sekuler, and D. Siemsen. 1983. Contrast sensitivity throughout adulthood. *Vision Research*, 1983.
- [36] M. S. El-Nasr and S. Yan. 2006. 3dVisual attention in 3D video games. *Workshop on CST*, 2006.
- [37] P. Orero and A. Remael. 2007. Media for all: Subtitling for the deaf, audio description, and sign language. *Rodopi*, 2007.
- [38] R. E. Nisbett and Y. Miyamoto. 2005. The influence of culture: holistic versus analytic perception. *Trends in Cognitive Sciences* 9, 10 (2005), 467–473.
- [39] E. Balçetis and G. D. Lassiter. 2010. *Social Psychology of Visual Perception*. Psychology Press, 2010.
- [40] A. Acar, T. Taura, E. Yamamoto, and N. F. M. Yusof. 2011. Object vs. relation: Understanding the link between culture and cognition with the help of wordnet. *Int. J. of Asian Lang. Proc* 21, 4 (2011), 199–208.
- [41] I. Wechsung, M. Schulz, K.-P. Engelbrecht, J. Niemann, and S. Möller. 2011. All users are (not) equal-the influence of user characteristics on perceived quality, modality choice and performance. In *Proceedings of the Paralinguistic Information and its Integration in Spoken Dialogue Systems Workshop*. Springer, New York, NY.

- [42] A. B. Naumann, I. Wechsung, and J. Hurtienne. 2010. Multimodal interaction: A suitable strategy for including older users? *Interacting with Computers* 22, 6 (2010), 465–474.
- [43] P. J. Silvia. 2008. Interest—The curious emotion. *Curr. Dir. Psychological Science* 17, 1 (2008), 57–60.
- [44] H. L. O'Brien and E. G. Toms. 2008. What is user engagement? A conceptual framework for defining user engagement with technology. *JAIST* 59, 6 (2008), 938–955.
- [45] P. Kortum and M. Sullivan. 2010. The effect of content desirability on subjective video quality ratings. *Human Factors: The Journal of the Human Factors and Ergonomics Society* 52, 1 (2010), 105–118.
- [46] J. Palhais, R. S. Cruz, and M. S. Nunes. 2011. Quality of experience assessment in internet TV. In *International Conference on Mobile Networks and Management*. Springer, Berlin, Heidelberg, 261–274.
- [47] A. Sackl, R. Schatz, and A. Raake. 2017. More than I ever wanted or just good enough? User expectations and subjective quality perception in the context of networked multimedia services. *QUEX* 2, 1 (2017), 3.
- [48] A. Sackl and R. Schatz. 2014. Got what you want? Modeling expectations to enhance web QoE prediction. *Quality of Multimedia Experience (QoMEX)*, IEEE, 2014.
- [49] A. Sackl and R. Schatz. 2014. Evaluating the influence of expectations, price and content selection on video quality perception. *Quality of Multimedia Experience (QoMEX)*, IEEE, 2014.
- [50] P. A. Kara, L. Bokor, A. Sackl, and M. Mourão. 2015. What your phone makes you see: Investigation of the effect of end-user devices on the assessment of perceived multimedia quality. *Quality of Multimedia Experience (QoMEX)*, IEEE, 2015.
- [51] P. G. Engeldrum. 2000. *Psychometric Scaling: A Toolkit for Imaging Systems Development*. Imcotek Press, 2000.
- [52] A. G. Greenwald, D. E. McGhee, and J. L. Schwartz. 1998. Measuring individual differences in implicit cognition: the implicit association test. *Journal of Personality and Social Psychology* 74, 6 (1998), 1464.
- [53] C. Keimel, J. Habigt, and K. Diepold. 2012. Challenges in crowd-based video quality assessment. *Quality of Multimedia Experience (QoMEX)*, IEEE, 2012.
- [54] T. Hofffeld, M. Hirth, J. Redi, F. Mazza, P. Korshunov, B. Naderi, M. Seufert, B. Gardlo, S. Egger, and C. Keimel. 2014. Best practices and recommendations for crowdsourced QoE-lessons learned from the qualinet task force. *Crowdsourcing* (2014).
- [55] M. Weisbuch, Z. Ivcevic, and N. Ambady. 2009. On being liked on the web and in the “real world”: Consistency in first impressions across personal webpages and spontaneous behavior. *JESP* 45, 3 (2009), 573–576.
- [56] M. Burke, C. Marlow, and T. Lento. 2009. Feed me: Motivating newcomer contribution in social network sites. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 2009.
- [57] H. Li, H. Wang, J. Liu, and K. Xu. 2012. Video sharing in online social networks: measurement and analysis. In *Proceedings of the International Workshop on Network and Operating System Support for Digital Audio and Video*. ACM, 2012.
- [58] L. T. Graham, C. J. Sandy, and S. D. Gosling. 2011. Manifestations of individual differences in physical and virtual environments. *The Wiley-Blackwell Handbook of Individual Differences* (2011), 773–800.
- [59] D. Garcia and S. Sikström. 2014. The dark side of facebook: Semantic representations of status updates predict the dark triad of personality. *Personality and Individual Differences* 67 (2014), 92–96.
- [60] Y. Bachrach, M. Kosinski, T. Graepel, P. Kohli, and D. Stillwell. 2012. Personality and patterns of Facebook usage. In *Proceedings of the 4th Annual ACM Web Science Conference*. ACM, 2012.
- [61] A. Eftekhari, C. Fullwood, and N. Morris. 2014. Capturing personality from facebook photos and photo-related activities: How much exposure do you need? *Computers in Human Behavior* 37 (2014), 162–170.
- [62] M. Kosinski, D. Stillwell, and T. Graepel. 2013. Private traits and attributes are predictable from digital records of human behavior. *Proceedings of the National Academy of Sciences* 110, 15 (2013), 5802–5805.
- [63] M. Park, M. Naaman, and J. Berger. 2016. A data-driven study of view duration on Youtube. *arXiv:1603.08308* (2016).
- [64] X. Yi, L. Hong, E. Zhong, N. N. Liu, and S. Rajan. 2014. Beyond clicks: Dwell time for personalization. In *Proceedings of the 8th ACM Conference on Recommender Systems*. ACM, 2014.
- [65] B. Gardlo, M. Ries, T. Hossfeld, and R. Schatz. 2012. Microworkers vs. facebook: The impact of crowdsourcing platform choice on experimental results. *Quality of Multimedia Experience (QoMEX)*, IEEE.
- [66] D. J. Stillwell and M. Kosinski. 2004. Mypersonality project: Example of successful utilization of online social networks for large-scale social research. *American Psychologist* 59, 2 (2004), 93–104.
- [67] H. A. Schwartz, J. Eichstaedt, and M. E. Seligman. 2013. Personality, gender, and age in the language of social media: The open-vocabulary approach. *PloS one* 8, 9 (2013), e73791.
- [68] D. Preotiu-Pietro, S. Volkova, V. Lampos, Y. Bachrach, and N. Aletras. 2015. Studying user income through language, behaviour and affect in social media. *PloS one* 10, 9 (2015), e0138717.
- [69] L. Zwarun and A. Hall. 2014. What's going on? Age, distraction, and multitasking during online survey taking. *Computers in Human Behavior* 41 (2014), 236–244.

- [70] S. R. Gulliver and G. Ghinea. 2006. Defining user perception of distributed multimedia quality. *ACM Transactions on Multimedia Computing, Communications, and Applications* 2, 4 (2006), 241–257.
- [71] S. D. Gosling, P. J. Rentfrow, and W. B. Swann. 2003. A very brief measure of the Big-Five personality domains. *Journal of Research in Personality* 37, 6 (2003), 504–528.
- [72] W. Robitza, M. N. Garcia, and A. Raake. 2015. At home in the lab: Assessing audiovisual quality of HTTP-based adaptive streaming with an immersive test paradigm. *Quality of Multimedia Experience (QoMEX)* IEEE, 2015.
- [73] A. Hanjalic and L.-Q. Xu. 2005. Affective video content representation and modeling. *TMM, IEEE* (2005).
- [74] D. Rao, D. Yarowsky, A. Shreevats, and M. Gupta. 2010. Classifying latent user attributes in Twitter. In *Proceedings of the 2nd International Workshop on Search and Mining User-Generated Contents*. ACM, 2010.
- [75] J. D. Burger, J. Henderson, G. Kim, and G. Zarrella. 2011. Discriminating gender on Twitter. *Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2011.
- [76] S. C. Guntuku, D. B. Yaden, M. L. Kern, L. H. Ungar, and J. C. Eichstaedt. 2017. Detecting depression and mental illness on social media: an integrative review. *CoBeHa* 18, 43–49.
- [77] S. C. Guntuku, W. Lin, M. J. Scott, and G. Ghinea. 2015. Modelling the influence of personality and culture on affect and enjoyment in multimedia. In *Proceedings of the 2015 International Conference on Affective Computing and Intelligent Interaction (ACII)*. IEEE, 236–242.
- [78] S. C. Guntuku, M. J. Scott, H. Yang, G. Ghinea, and W. Lin. 2015. The CP-QAE-I: A video dataset for exploring the effect of personality and culture on perceived quality and affect in multimedia. In *Proceedings of the 2015 7th International Workshop on Quality of Multimedia Experience (QoMEX)*. IEEE, 1–7.
- [79] S. C. Guntuku, J. R. Ramsay, R. M. Merchant, and L. H. Ungar. 2017. Language of ADHD in adults on social media. *Journal of Attention Disorders*, 1087054717738083.
- [80] S. C. Guntuku, W. Lin, J. Carpenter, W. K. Ng, L. H. Ungar, and D. Preoțiu-Pietro. 2017. Studying personality through the content of posted and liked images on Twitter. In *Proceedings of the 2017 ACM on Web Science Conference*. ACM, 223–227.
- [81] S. C. Guntuku, J. T. Zhou, S. Roy, L. Weisi, and I. W. Tsang. 2016. Who likes what, and why? Insights into personality modeling based on imagelikes. *IEEE Transactions on Affective Computing*.
- [82] S. C. Guntuku, S. Roy, W. Lin, K. Ng, N. W. Keong, and V. Jakhethiya. 2016. Personalizing user interfaces for improving quality of experience in VoD recommender systems. In *Proceedings of the 2016 8th International Conference on Quality of Multimedia Experience (QoMEX)*. IEEE, 1–6.
- [83] S. C. Guntuku, L. Qiu, S. Roy, W. Lin, and V. Jakhethiya. 2015. Do others perceive you as you want them to?: Modeling personality based on selfies. In *Proceedings of the 1st International Workshop on Affect & Sentiment In Multimedia*. ACM, 21–26.
- [84] M. J. Scott, S. C. Guntuku, W. Lin, and G. Ghinea. 2016. Do personality and culture influence perceived video quality and enjoyment? *IEEE Transactions on Multimedia* 18, 9, 1796–1807.
- [85] G. Ghinea and S. Y. Chen. 2006. Perceived quality of multimedia educational content: A cognitive style approach. *Multimedia Systems* 11, 3, 271–279.
- [86] S. R. Gulliver and G. Ghinea. 2010. Cognitive style and personality: Impact on multimedia perception. *Online Information Review* 34, 1, 39–58.
- [87] S. C. Guntuku, M. J. Scott, G. Ghinea, and W. Lin. 2016. Personality, culture, and system factors - Impact on affective response to multimedia. *arXiv Preprint arXiv:1606.06873*.

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